ENR 1.5 HOLDING, APPROACH AND DEPARTURE PROCEDURES

1 GENERAL

The holding, approach and departure procedures in use are based on those contained in the latest edition of ICAO Doc 8168- OPS/611 (PANS-OPS).The holding and approach procedures in use have been based on the values and factors contained in part 11 of the PANS-OPS.

The ILS Outer Marker (OM) is a compulsory reporting point for all aircraft on an ILS approach. In the event that the ILS OM has been replaced with a DME distance fix the DME distance fix will be used as the OM and reporting done accordingly.

1.1 Provisions

a) Aircraft must be able to receive the ATIS.
b) SIDS and STARS are announced in operation on ATIS.
c) The active runway is announced on the ATIS.
d) SIDS and STARS will only be in force when Surveillance Radar in operation.
e) If unable to comply with SID or STAR, notify ATC.
f) If unable to fly TRACKS, notify ATC.
g) Pilots wishing to practice a CAT II ILS approach or a GNSS approach must either request it on the flight plan in field 18, request it at least one (1) HR before by phone, or by radio at 100NM DME JSV TGV or CTV.

2 ARRIVING FLIGHTS

IFR flights entering and landing within controlled airspace will be cleared to a specific holding point and instructed to contact the Approach Control at a specified time, level or position.

The terms of this clearance shall be adhered to until further instructions are received from Approach Control.

If the clearance is reached before further instructions have been received, holding procedures shall be carried out at the level cleared.

1) If it becomes apparent that delays will be encountered by arriving aircraft, the operator or his designated representative shall be notified and kept informed of any changes in such delays.This is to facilitate possible timely diversionary action.
2) To expedite departing traffic, arriving aircraft may be required to report when leaving or passing a reporting point, or when starting a procedure turn or base turn, or to provide other information as requested by ATC.

3) An IFR flight shall not be cleared to descend below the appropriate initial approach altitude as specified for a particular instrument approach unless:

a) The pilot has reported passing an appropriate point defined by a radio aid; or

b) The pilot reports that he/she has and can maintain visual contact with the aerodrome; or

c) The pilot is conducting a visual approach; (The limit for a visual approach is 25NM from the destination aerodrome); or

d) The pilot is conducting an approach under RADAR vectors and control

4) All inbound aircraft to O R Tambo (Johannesburg), Lanseria, Chief Dawid Stuurman (Port Elizabeth), Bram Fischer (Bloemfontein), King Phalo (East London), King Shaka (Durban), Cape Town Airports not on a STAR, will comply with the following speed restrictions unless advised otherwise by ATC:

a) Within 50DME JSV/PEV/CTV/TGV/BLV/ELV: 250KIAS or less

b) Within 15DME JSV/PEV/CTV/TGV/BLV/ELV: 210KIAS or less

c) Speed limit points (SLP) will be defined per STAR procedure.

d) In addition, all aircraft at or below FL 100 will fly a speed not exceeding 250 KIAS. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency pilots are not to request cancellation of speed restrictions.

2.1 VMC AND VISUAL APPROACHES

The purpose is to define the two types of approach clearances and pilots are requested to note the essential differences.

2.2 VMC APPROACH

When requested by the pilot and if so prescribed by the ATSU, an arriving aircraft may be cleared to descend below the initial approach ALT subject to maintaining by day;

a) own separation;

b) VMC;

c) visual contact with the aerodrome; and

d) by night - provided that there is no reduction of standard separation involved.

2.3 VISUAL APPROACH

An IFR flight may be cleared to execute a visual approach provided that the pilot can maintain visual reference to the terrain and;

a) the reported ceiling is at or above the approved initial approach ALT for the aircraft so cleared;

b) reasonable assurance exists that a visual approach and landing can be completed;

c) be within 25 NM of the destination aerodrome.

2.3.1 Separation shall be provided between an aircraft cleared to execute a visual approach and other traffic within controlled airspace.

2.4 MANDATORY RADIO COMMUNICATIONS

2.4.1 Radio communication failure (RCF) procedures - General

1) When an aircraft fails to establish contact with the aeronautical station on the designated frequency, it shall attempt to establish contact on another frequency appropriate to the route. If this attempt fails, the aircraft shall attempt to establish communication with another aircraft or other aeronautical stations on frequencies appropriate to the route. In addition, an aircraft shall monitor the appropriate VHF frequency for calls from nearby aircraft or aeronautical stations.
2) If these attempts fail, the aircraft station shall continue to transmit position reports and its intentions as appropriate on the designated frequencies, preceded by the phrase Transmitting blind. Such messages shall be transmitted twice and, if necessary, include the addressee (s) for which the message is intended.
3) If no communication is received or other indication that one-way communications are possible, the aircraft shall set its transponder to Code 7600 and proceed with the lost communications procedures.
4) In any case, whereby an aircraft having suffered a communication failure in flight arrives at an aerodrome, it shall keep watch for such instructions as may be issued by visual signals for the aerodrome control tower or other facility.

2.4.2 RCF procedures - VFR

a) If the communications failure occurs while operating in accordance with VFR, the aircraft shall continue to fly in visual meteorological conditions (VMC) and land at the nearest suitable aerodrome using the standard RCF arrival procedures prescribed in Appendix 1 to this section; or if other procedures have been published by the Director for a specific aerodrome, in accordance with such procedures.
b) The operator shall report its arrival by the most expeditious means to the appropriate air traffic services unit (ATSU).

2.4.3 RCF procedures - IFR

1) If the communications failure occurs while operating in accordance with IFR and VMC are encountered, the aircraft shall -
   a) continue to fly in VMC; land at the nearest suitable aerodrome in accordance with -
   i) the standard RCF arrival procedures prescribed in Appendix 1 to this section; or
   ii) if other procedures have been published by the Director for a specific aerodrome, in accordance with such procedures, and
   b) report its arrival by the most expeditious means to the appropriate ATSU; or
   c) if unable to ensure VMC conditions exist to a suitable aerodrome, complete an IFR flight in accordance with paragraph 2 below.

2) If the communications failure occurs while operating in accordance with IFR while in IMC or, if in VMC but unable to maintain VMC, the aircraft shall -
   a) in airspace where an ATS surveillance system is not used in the provision of air traffic control, maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 20 minutes following the aircraft's failure to report its position over a compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan;
   b) in airspace where an ATS surveillance system is used in the provision of air traffic control, maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 7 minutes following -
   i) the time the last assigned level or minimum flight altitude was reached;
   ii) the time the transponder was set to Code 7600; or
   iii) the aircraft's failure to report its position over a compulsory reporting point, whichever is later; and thereafter adjust level and speed in accordance with the filed flight plan;
   c) when being radar vectored or having been directed by ATC to proceed offset using area navigation (RNAV) without a specified limit, rejoin the current flight plan route no later than the next significant point, taking into consideration the applicable minimum flight altitude;
   d) proceed according to the current flight plan route to the appropriate designated navigation aid or fix serving the destination aerodrome and, when required to ensure compliance with subparagraph (e), hold over this aid or fix until commencement of descent;
   e) commence descent from the navigation aid or fix specified in subparagraph (d) at, or as close as possible to, the expected approach time last received and acknowledged or, if no expected approach time has been received and acknowledged, at, or as close as possible to, the estimated time of arrival resulting from the current flight plan;
   f) complete a normal instrument approach procedure as specified for the designated navigation aid or fix; and
   g) land, if possible, within 30 minutes after the estimated time of arrival specified in (e) or the last acknowledged expected approach time, whichever is later.

APPENDIX 1

Standard Radio Communications Failure Procedure - VFR Arrivals.

1) Maintain squawk of 7600.
2) Make a relevant blind broadcast to traffic in the area advising of the probability of a radio communication failure, position and intentions.
3) Select landing lights on.
4) Approaching the aerodrome, make a relevant blind broadcast to traffic on the controlled airfields frequency to indicate the probability of a radio communication failure, position and intentions.
5) Join overhead the aerodrome at a height of 1000 feet above circuit altitude to ascertain which is the active runway in use.
6) Conform to the circuit pattern while joining, preferably on the downwing leg.
7) Continue to make blind broadcasts on the controlled airfields frequency to indicate the position in the circuit pattern.
8) Land and vacate the runway expeditiously and safely.
9) Taxi to the nearest parking area and shutdown.
10) Inform the owner or operator and ATC.
11) Make the relevant entry in the aircraft's flight folio.

Note: This procedure is to be used in the event the aerodrome at which the landing is to take place does not have specific procedures to be followed as published in the AIP

3 CAPE TOWN FIR

Cape Town International Airport.

STAR Applicable only when instructed by ATC or announced on ATIS.

3.1 STANDARD TERMINAL ARRIVAL ROUTES (STAR) RWY01/19.

Applicable to the following aircraft:

1) Aircraft already cleared for the STAR.
2) All aircraft inside the Cape Town FIR with flight plan destination as Cape Town International Airport or Ysterplaat Military, and able to comply with the STAR.
3) When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.
4) Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency, pilots are not to request cancellation of speed restrictions
5) Communication Failure Procedure for the STAR

a) Applicable to all IFR aircraft inbound to FACT irrespective of IMC or VMC
b) If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL100, whichever is the higher. Comply with the Communication Failure procedure associated with that STAR position.

6) Missed Approach: Missed Approaches with the intention of carrying out another approach should follow the procedural Missed Approach instructions unless advised differently by ATC. In the event of a missed approach with the intention of diverting to an alternate aerodrome proceed as follows:

i) Diverting to the East follow the routing for the TETAN ONE CHARLIE DEPARTURE
ii) Diverting to the South-East, follow the routing for the OKTED ONE ALFA DEPARTURE.
iii) Diverting to the North, follow the routing for the KODES ONE ALFA DEPARTURE.

7) Provisions

a) Aircraft must be able to receive the ATIS.
b) SIDS and STARS are announced in operation on ATIS.
c) Runway 01 is in use on the ATIS.
d) SIDS and STARS will only be in force when Surveillance Radar in operation.
e) If unable to comply with SID or STAR, notify ATC.

RUNWAY 01

Clearance	Details
ERDAS 1A ARRIVAL (ERDAS 1A)	
ERDAS 1A ARRIVAL (ERDAS 1A) Note: ERDAS at position:S332450.10 E0191112.85 (R-065/44.2DME "CTV")	Leave ERDAS on R-065 "CTV" inbound. Passing 8 DME "CTV" turn left to track 190°M for radar vectoring onto the ILS RWY 01. Pass 30DME "CTV" maintaining 250 KT IAS or less. At 8 DME "CTV" maintaining 210 KT IAS or less. Once established on the ILS Localizer maintain 180 KT IAS until 10 DME. Passing OM 150 KT IAS or less.
Communication Failure (ERDAS 1A) (Squawk 7600)	
Before ERDAS: Proceed to ERDAS and enter the ERDAS hold. Hold at last assigned level for minimum 5 minutes, then climb/descend to FL100 in the hold. Leave ERDAS at FL100 and continue on the "After ERDAS" communication failure procedure. After ERDAS: Continue on the routing for the ERDAS 1A STAR and maintain last assigned level. Crossing R-156 "CTV" on track 190°M turn right direct "CTV", then climb/descend to 6500 FT ALT. On reaching "CTV" complete the VOR/DME/ILS approach and land RWY 01. Note: Routing (including MNM Noise Routing). To avoid high ground, when established on track 190°M and below 5700 FT ALT, do not exceed 16DME "CTV".	
GETEN 1A ARRIVAL (GETEN 1A)	
GETEN 1A ARRIVAL (GETEN 1A) NOTE: GETEN at position:S340326.38 E0193611.70 (R-120/50.1DME "CTV")	Leave GETEN on R-120 "CTV". Passing 9DME "CTV" turn left to track 190°M for radar vectoring onto the ILS RWY 01. Pass 40DME "CTV" maintaining 250 KT IAS or less. Pass 9 DME "CTV" maintaining 210KT IAS or less. Once established on the ILS Localizer maintain MAXIMUM of 180 KT IAS until 10 DME. Passing OM 150 KT IAS or less.
Communication Failure (GETEN 1A) (Squawk 7600)	
Before GETEN: Proceed to GETEN and enter the GETEN hold. Hold at last assigned level for minimum 5 minutes, then climb/descend to FL100 in the hold. Leave GETEN at FL100 and continue on the "After GETEN" communication failure procedure. After GETEN: Continue on the routing for the GETEN 1A STAR and maintain last assigned level. Crossing R-156 "CTV" on track 190°M turn right direct "CTV" then descend to 6500FT ALT. On reaching "CTV" complete the VOR/DME/ILS APCH and land RWY 01. Note: Routing (including MNM Noise Routing). Due to terrain do not cross (west of) R-210 "CTV", and to avoid high ground, when established on track 190°M and below 5700 FT ALT, do not exceed 16DME "CTV".	
ASPIK 1A ARRIVAL (ASPIK1A)	
ASPIK 1A ARRIVAL (ASPIK 1A) Note: Significant point ASPIK at position: S340618.3199 E 0180800.8068 (R-275/25DME "CTV")	Leave ASPIK on R-275 "CTV" (Inbound). Passing 8 DME "CTV" turn right to track 190°M for radar vectoring onto the ILS RWY 01. Pass 30DME “CTV” maintaining 250KT IAS. Cross ASPIK maintaining 210 KT IAS or less. Once established on the ILS Localizer maintain MAXIMUM of 180 KT IAS until 10 DME and to cross the Outer Marker at 150 KT IAS.

Communication Failure (ASPIK 1A)(Squawk 7600)	
Before ASPIK. Proceed to ASPIK and enter the ASPIK hold. Hold at last assigned level for MINIMUM 5 minutes, then descend to 6500 FT ALT in the hold. Leave ASPIK on the 'after ASPIK' communication failure procedure. After "ASPIK". Continue on the routing for the ASPIK 1A STAR and maintain 6500 FT ALT. Crossing R-225 “CTV” on track 190°M turn left direct “CTV”. On reaching “CTV” complete the VOR/DME/ILS approach and land RWY 01. Note: Due to terrain do not cross (east of) R-180 “CTV”.	
ERDAS 1C ARRIVAL (ERDAS1C)	
ERDAS 1C ARRIVAL (ERDAS 1C) NOTE: ERDAS at position:S332450.10 E0191112.85 (R-065/44.2DME “CTV”)	To CT830, to CT831, to CT832, to CT833 at FL100 or above, to UTIRO at 6500FT or above. At UTIRO, continue on the RNAV (GNSS) APCH RWY 01.
Communication Failure (ERDAS 1C)(Squawk 7600)	
Before ERDAS: Proceed to ERDAS and enter the ERDAS RNAV (GNSS) hold at last assigned level. Descend to FL150 in the hold. Leave ERDAS on the “After ERDAS” Communication Failure Procedure. After ERDAS: Continue on the ERDAS 1C RNAV (GNSS) STAR. At UTIRO, continue on the RNAV (GNSS) APCH RWY 01 Communication Failure Procedure.	
ASPIK 1C ARRIVAL (ASPIK1C)	
ASPIK 1C ARRIVAL (ASPIK 1C) Note: Significant point ASPIK at position: S340618.3199 E0180800.8068 (R-275/25DME “CTV”)	To CT811 at or above FL100, turn right to EKBEV at or above 6100FT. At EKBEV continue on the RNAV (GNSS) APCH RWY 01.
Communication Failure (ASPIK 1C)(Squawk 7600)	
Before ASPIK: Proceed to ASPIK and enter the ASPIK 1C RNAV (GNSS) hold at last assigned level. Descend to FL100 in the hold. Leave ASPIK on the “After ASPIK” Communication Failure Procedure. After ASPIK: Continue on the ASPIK 1C RNAV (GNSS) STAR. At EKBEV, continue on the RNAV (GNSS) APCH RWY 01 Communication Failure Procedure.	
GETEN 1C ARRIVAL (GETEN 1C)	
GETEN 1C ARRIVAL (GETEN 1C) NOTE: GETEN at position:S340326.38 E0193611.70 (R-120/50.1DME "CTV")	To CT812, to CT813 at FL100 or above, turn left to CT814, turn left to UTIRO at 6500FT ALT or above. At UTIRO, continue on the RNAV (GNSS) APCH RWY 01.
Communication Failure (GETEN 1C)(Squawk 7600)	
Before GETEN: Proceed to GETEN and enter the GETEN RNAV (GNSS) hold at last assigned level. Descend to FL150 in the hold. Leave GETEN on the “After GETEN” Communication Failure Procedure. After GETEN: Continue on the GETEN 1C RNAV (GNSS) STAR. At UTIRO, continue on the RNAV (GNSS) APCH RWY 01 Communication Failure Procedure.	

2. Standard Terminal Arrival Routes (STAR) RWY 19

Applicable to the following aircraft

2.1.Aircraft already cleared for the STAR.

2.2.All aircraft inside the Cape Town FIR with flight plan destinations as Cape Town International or Ysterplaat Military, and able to comply with a STAR.

2.3.When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.

2.4.Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency pilots are not to request cancellation of speed restrictions.

2.5.Communication Failure Procedures for STAR
If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL100, whichever is the highest. Comply with the Communication Failure procedure associated with that STAR position.

2.6.Missed Approach
Missed Approaches with the intention of carrying out another approach should follow the procedural Missed Approach instructions unless advised differently by ATC. In the event of a missed approach with the intention of diverting to an alternate aerodrome proceed as follows:

a.Diverting to the east follow the routing for the TETAN ONE BRAVO DEPARTURE

b.Diverting to the south-east, follow the routing for the OKTED ONE BRAVO DEPARTURE

c.Diverting to the north and north-west, follow the routing for the KODES ONE BRAVO DEPARTURE

2.7Provisions

a.Aircraft must be able to receive the ATIS.

b.SIDS and STARS are announced in operation on ATIS.

c.Runway 19 is in use on the ATIS.

d.SIDS and STARS will only be in force when Surveillance Radar in operation.

e.If unable to comply with SID or STAR, notify ATC.

RUNWAY19

Clearance	Details
ERDAS 1B ARRIVAL (ERDAS 1B)	
ERDAS 1B ARRIVAL (ERDAS 1B) NOTE: ERDAS at position: S332450.10 E0191112.85 (R-065/44.2DME "CTV")	Leave ERDAS on R-065 "CTV" inbound. Passing 11 DME "CTV" turn right to track 010°M for radar vectoring onto the ILS RWY 19. Pass 30DME "CTV" maintaining 250 KIAS or less. Pass 11 DME "CTV" maintaining 210 KIAS or less. Once established on the ILS Localizer maintain MAXIMUM of 180 KT IAS until 10 DME. Passing Outer Marker 150 KIAS or less.
Communication Failure (ERDAS 1B)(Squawk 7600)	
Before ERDAS: Proceed to ERDAS and enter the ERDAS hold. Hold at last assigned level for minimum 5 minutes, then climb/descend to FL100 in the hold. Leave ERDAS at FL100 and continue on the "After ERDAS" communication failure procedure. After ERDAS: Continue on the routing for the ERDAS 1B STAR and maintain last assigned level. Crossing R-035 "CTV" on track 010°M turn left direct "CTV" then descend to 6500FT ALT. On reaching "CTV" complete the VOR/DME/ILS approach and land RWY 19.	
GETEN 1B ARRIVAL (GETEN 1B)	
GETEN 1B ARRIVAL (GETEN 1B) NOTE: GETEN at position: S340326.38 E0193611.70 (R-120/50.1DME "CTV")	Leave GETEN on R-120 "CTV". Passing 8DME "CTV" turn right to track 010°M for radar vectoring onto the ILS RWY 19. Pass 30DME "CTV" maintaining 250 KIAS or less. Pass 8DME "CTV" maintaining 210 KIAS or less. Once established on the ILS Localizer maintain MAXIMUM of 180 KT IAS until 10 DME. Passing Outer Marker 150 KIAS or less.
Communication Failure (GETEN 1B)(Squawk 7600)	
Before GETEN: Proceed to GETEN and enter the GETEN hold. Hold at last assigned level for minimum 5 minutes then climb/descend to FL100. Leave GETEN at FL100 and continue on the "After GETEN" communications failure procedure. After GETEN: Continue on the routing for the GETEN 1B STAR, maintain last assigned level. Crossing R-035 "CTV" on track 010°M turn left direct "CTV" then descend to 6500FT ALT. On reaching "CTV" complete the VOR/DME/ILS approach and land RWY 19.	
ASPIK 1B ARRIVAL (ASPIK 1B)	
ASPIK 1B ARRIVAL (ASPIK 1B) NOTE: Significant point ASPIK at position: S340618.3199 E 0180800.8068 (R-275/25DME "CTV")	Leave ASPIK on R-275 "CTV" (inbound). Passing 8DME "CTV" turn left to track 010°M for radar vectoring onto the ILS RWY 19. Pass 30DME "CTV" maintaining 250 KIAS or less. Pass ASPIK maintaining 210 KIAS or less. Once established on the ILS Localizer maintain MAXIMUM of 180 KIAS until 10 DME and to cross the Outer Marker at 150 KIAS.
Communication Failure (ASPIK 1B)(Squawk 7600)	
Before ASPIK. Proceed to ASPIK and enter the ASPIK hold. Hold at last assigned level for MINIMUM 5 minutes, then descend to 6500 FT ALT in the hold. Leave ASPIK on the 'after ASPIK' communication failure procedure After "ASPIK". Continue on the routing for the ASPIK 1B STAR, maintain last assigned level. Crossing R-320 "CTV" on track 010°M turn right direct "CTV" and descend to 6500 FT ALT. On reaching "CTV" complete the VOR/DME/ILS approach and land RWY 19.	

4 JOHANNESBURG FIR:

King Shaka International Airport

All STARs applicable for runway 06. Use STARS for runway 06 only when instructed by ATC or announced on ATIS.

4.1 STANDARD TERMINAL ARRIVAL ROUTES (STAR) RWY06

4.1.1 Applicable to the following aircraft:

a) Aircraft already cleared for the STAR;

b) All aircraft inside the Johannesburg FIR with flight plan destination as King Shaka International and able to comply with the STAR.

c) When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.

d) Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency, pilots are not to request cancellation of speed restrictions.

4.1.2 Communication Failure Procedures for the STAR.

If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL130, whichever is the highest. Comply with the Communication Failure procedure associated with that STAR position.

4.1.3 Missed Approach

Missed Approaches with the intention of carrying out another approach should follow the procedural Missed Approach instructions unless advised differently by ATC. In the event of a missed approach with the intention of diverting to an alternate aerodrome proceed as follows:

a) Comply with the ITMIL 1C RNAV (GNSS) SID. At ITMIL continue to the diversion aerodrome.

4.1.4 Provisions

a) If unable to comply with SID or STAR, notify ATC

b) Aircraft must be able to receive the ATIS.

c) SIDs and STARs are announced in operation on ATIS.

d) Runway 06 is in use on the ATIS.

e) SIDs and STARs will only be in force when Surveillance Radar in operation.

f) General Aviation Traffic operates below the TMA up to 3000 FT ALT and is considered separated from traffic operating in the TMA.

RUNWAY 06

Clearance	Details
APMAT ONE ALPHA RNAV (GNSS) ARRIVAL (APMAT 1A)	
APMAT 1A RNAV (GNSS) ARRIVAL Note: Significant point APMAT at position: S284723.78 E0305659.63 (R014/50 DME TGV)	Leave APMAT for LE002. At LE002 turn right for LE1N1. At LE1N1 turn left for radar vectors onto ILS LOC RWY 06. Pass APMAT 250KT IAS or less. Pass LE001 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC
Communication Failure (APMAT 1A)(Squawk 7600)	
Before APMAT Proceed to APMAT and enter the APMAT RNAV (GNSS) hold. Hold at last assigned level, for minimum 5 MIN, then descend to FL130 in the hold or maintain last assigned level if below FL130. Leave APMAT on the "After APMAT" Communication Failure Procedure. After APMAT Continue on the APMAT 1A RNAV (GNSS) STAR maintaining last assigned level. Passing LE001 descend to FL080. At LE002 (not below 6300' ALT) turn right to LE1N1 (not below 5600' ALT). Continue with the RNAV (GNSS) RWY 06 to LE1N1 and intercept the ILS LOC RWY06 and land RWY 06. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the ITMIL 1C RNAV (GNSS) SID. At ITMIL continue to the diversion aerodrome.	
DUNSA ONE ALPHA RNAV (GNSS) ARRIVAL (DUNSA 1A)	
DUNSA 1A RNAV (GNSS) ARRIVAL Note: Significant point DUNSA at position: S301406.71 E0302914.71 (R246/50 DME TGV)	Leave DUNSA for LE1N2. At LE1N2 turn left to LE1T1 and intercept the ILS LOC RWY 06. Pass DUNSA 250KT IAS or less. Pass LE005 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC.
Communication Failure (DUNSA 1A)(Squawk 7600)	
Before DUNSA Proceed to DUNSA and comply with the “After DUNSA” communication failure procedure. After DUNSA Continue on the DUNSA 1A RNAV (GNSS) STAR maintaining last assigned level. Passing LE005 descend to FL080. At LE1N2 (not below 5100' ALT) continue to LE1T1 (not below 4000' ALT). Continue with the RNAV (GNSS) RWY 06 APCH and intercept the ILS LOC RWY 06 and land RWY 06. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the ITMIL 1C RNAV (GNSS) SID. At ITMIL continue to the diversion aerodrome.	
GETOK ONE CHARLIE RNAV (GNSS) ARRIVAL (GETOK 1C)	
GETOK 1C RNAV (GNSS) ARRIVAL Note: Significant point GETOK at position: S293046.69 E0300917.83 (R301/51 DME TGV)	Leave GETOK for LE010. At LE010 proceed to LE1N2. At LE1N2 turn left to LE1T1 and intercept the ILS LOC RWY 06. Pass GETOK 250KT IAS or less. Pass LE009 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC.

Clearance	Details
Communication Failure (GETOK 1C)(Squawk 7600)	
Before GETOK Proceed to GETOK and enter the GETOK RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL120 in the hold or maintain last assigned level if below FL120. Leave GETOK on the "After GETOK" Communication Failure Procedure. After GETOK Continue on the GETOK 1C RNAV (GNSS) STAR, maintaining FL120. At LE010 (not below 8000' ALT) continue to LE1N2. At LE1N2 (not below 5500' ALT) turn left to LE1T1 (not below 4000' ALT) and continue with the RNAV (GNSS) RWY 06 APCH and intercept the ILS LOC RWY 06 and land RWY 06. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the ITMIL 1C RNAV (GNSS) SID. At ITMIL continue to the diversion aerodrome.	
ITMIL ONE ALPHA RNAV (GNSS) ARRIVAL (ITMIL 1A)	
ITMIL 1A RNAV (GNSS) ARRIVAL Note: Significant point ITMIL at position: S290151.05 E0314838.40 (R070/50 DME TGV)	Leave ITMIL for LE006. At LE006 proceed to LE013. At LE013 turn right to LE1N3 for radar vectors onto the ILS LOC RWY 06. Pass ITMIL 250KT IAS or less. Pass LE006 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS, or as cleared by ATC
Communication Failure (ITMIL 1A)(Squawk 7600)	
Before ITMIL Proceed to ITMIL and enter the ITMIL RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL090 in the hold or maintain last assigned level if below FL090. Leave ITMIL on the "After ITMIL" Communication Failure Procedure. After ITMIL Continue on the ITMIL 1A RNAV (GNSS) STAR, maintaining last assigned level. At LE013 descend to FL080 and continue to LE1N3. At LE1N3 (not below 5600' ALT) continue with the RNAV (GNSS) RWY 06 APCH and intercept the ILS LOC RWY 06 and land RWY 06. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the ITMIL 1C RNAV (GNSS) SID. At ITMIL continue to the diversion aerodrome.	

King Shaka International Airport

All STARs applicable for runway 24. Use STARs for runway 24 only when instructed by ATC or announced on ATIS.

4.2 STANDARD TERMINAL ARRIVAL ROUTES (STAR) RWY24

4.2.1 Applicable to the following aircraft:

a) Aircraft already cleared for the STAR;

b) All aircraft inside the Johannesburg FIR with flight plan destinations as King Shaka International, and able to comply with a STAR.

c) When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.

d) Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency, pilots are not to request cancellation of speed restrictions.

4.2.2 Communication Failure Procedures for STAR.

If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL130, whichever is the highest. Comply with the Communication Failure procedure associated with that STAR position.

4.2.3 Missed Approach

Missed Approaches with the intention of carrying out another approach should follow the procedural Missed Approach instructions unless advised differently by ATC. In the event of a missed approach with the intention of diverting to an alternate aerodrome proceed as follows:

a) Comply with the OKTAN 1A RNAV (GNSS) SID. At OKTAN continue to the diversion aerodrome.

4.2.4 Provisions

a) If unable to comply with SID or STAR, notify ATC
b) Aircraft must be able to receive the ATIS.
c) SIDs and STARs are announced in operation on ATIS.
d) Runway 24 is in use on the ATIS.
e) SIDs and STARs will only be in force when Surveillance Radar in operation.
f) General Aviation Traffic operates below the TMA up to 3000 FT ALT and is considered separated from traffic operating in the TMA.

RUNWAY 24

Clearance	Details
APMAT ONE BRAVO RNAV (GNSS) ARRIVAL (APMAT 1B)	
APMAT 1B RNAV (GNSS) ARRIVAL Note: Significant point APMAT at position S284723.78 E0305659.63 (R014/50 DME TGV)	Leave APMAT for LE004. At LE004 proceed to LE2N2. At LE2N2 turn right to LE2T1 and intercept the ILS LOC RWY 24. Pass APMAT 250KT IAS or less. Pass LE003 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS, or as cleared by ATC.
Communication Failure (APMAT 1B)(Squawk 7600)	
Before APMAT Proceed to APMAT and enter the APMAT RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN then descend to FL130 in the hold or maintain last assigned level if below FL130. Leave APMAT on the "After APMAT" Communication Failure Procedure. After APMAT Continue on APMAT 1B RNAV (GNSS) STAR, maintaining last assigned level. At LE003 descend to FL080 and proceed to LE004. At LE004 (not below 6500' ALT) proceed to LE2N2. At LE2N2 (not below 6100' ALT) turn right to LE2T1 and continue with the RNAV (GNSS) RWY 24 APCH and intercept the ILS LOC RWY 24 and land RWY 24. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the OKTAN 1A RNAV (GNSS) SID. At OKTAN continue to the diversion aerodrome.	
DUNSA ONE BRAVO RNAV (GNSS) ARRIVAL (DUNSA1B)	
DUNSA 1B RNAV (GNSS) ARRIVAL Note: Significant point DUNSA at position: S301406.71 E0302914.71 (R246/50 DME TGV)	Leave DUNSA for LE008. At LE008 turn left proceeding to LE2N1 for radar vectors onto the ILS LOC RWY 24 and land. Pass DUNSA 250KT IAS or less. Pass LE007 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC.
Communication Failure (DUNSA 1B)(Squawk 7600)	
Before DUNSA Proceed to DUNSA and comply with the After DUNSA Communication Failure Procedure. After DUNSA Continue on the DUNSA 1B RNAV (GNSS) STAR and maintain last assigned level. Passing LE007 descend to FL080. At LE008 (not below 5600' ALT), turn left to LE2N1 and continue with the RNAV (GNSS) RWY 24 APCH and intercept the ILS LOC RWY 24 and land RWY 24. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the OKTAN 1A RNAV (GNSS) SID. At OKTAN continue to the diversion aerodrome.	
GETOK ONE DELTA RNAV (GNSS) ARRIVAL (GETOK 1D)	
GETOK 1D RNAV (GNSS) ARRIVAL Note: Significant point GETOK at position: S293046.69 E0300917.83 (R301/51 DME TGV)	Leave GETOK for LE012. At LE012 turn left to LE2N3 for radar vectors onto the ILS LOC RWY 24. Pass GETOK 250KT IAS or less. Pass LE011 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC.

Clearance	Details
Communication Failure (GETOK 1D)(Squawk 7600)	
Before GETOK Proceed to GETOK and enter the GETOK RNAV (GNSS)hold. Hold at last assigned level for minimum 5 MIN, then descend to FL120 in the hold, or maintain last assigned level if below FL120. Leave GETOK on the "After GETOK" Communication Failure Procedure. After GETOK Continue on the GETOK 1D RNAV (GNSS) STAR and maintain FL120. At LE012 (not below 7500'ALT) turn left to LE2N3. At LE2N3 (not below 5600' ALT) and continue with the RNAV (GNSS) RWY 24 APCH and intercept the ILS LOC RWY 24 and land RWY 24. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the OKTAN 1A RNAV (GNSS) SID. At OKTAN continue to the diversion aerodrome.	
ITMIL ONE BRAVO RNAV (GNSS) ARRIVAL (ITMIL 1B)	
ITMIL 1B RNAV (GNSS) ARRIVAL Note: Significant point ITMIL at position: S290151.05 E0314838.40 (R070/50 DME TGV)	Leave ITMIL for LE2N2. At LE2N2 turn left and intercept the ILS LOC RWY 24. Pass ITMIL 250KT IAS or less. Pass LE014 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing 4 DME 150KT IAS or as cleared by ATC.
Communication Failure (ITMIL 1B)(Squawk 7600)	
Before ITMIL Proceed to ITMIL and enter the ITMIL RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL090 in the hold or maintain last assigned level if below FL090. Leave ITMIL on the "After ITMIL" Communication Failure Procedure. After ITMIL Continue on the ITMIL 1B RNAV (GNSS) STAR and maintain last assigned level. At LE014 descend to FL080. At LE2N2 (not below 5600' ALT) turn left and continue with the RNAV (GNSS) APCH and intercept the ILS LOC RWY 24 and land RWY 24. In the event of a missed approach with the intention of diverting to an alternate aerodrome, proceed as follows: Comply with the OKTAN 1A RNAV (GNSS) SID. At OKTAN continue to the diversion aerodrome.	

5 JOHANNESBURG FIR

O R Tambo International Airport

All STAR applicable for runway 03R. Use STAR for runway 03L only when instructed by ATC or announced on ATIS.

5.1 Standard Terminal Arrival Routes (STAR) RWY 03

5.1.1 Applicable to the following aircraft:

a) Aircraft already cleared for the STAR.

b) All aircraft inside the Johannesburg FIR with flight plan destinations as O R Tambo International or Lanseria International and able to comply with a STAR.

c) When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.

d) Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency, pilots are not to request cancellation of speed restrictions.

5.1.2 Communication Failure procedures for STAR.

a) Applicable to all IFR aircraft inbound to FAOR irrespective of IMC or VMC.

b) If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL090, whichever is the highest. Comply with the Communication Failure procedure associated with that STAR.

5.1.3 Provisions

a) Aircraft must be able to receive the ATIS.

b) SIDs and STARs are announced in operation on ATIS.

c) Runway 03 is in use on the ATIS.

d) SIDs and STARs will only be in force when surveillance radar in operation.

e) If unable to comply with SID or STAR, notify ATC

f) General Aviation traffic operates below the TMA up to 7500FT ALT and is considered separated from traffic operating in the TMA

RUNWAY 03

Clearance	Details
AVILO ONE ALPHA ARRIVAL (AVILO 1A)	
AVILO 1A ARRIVAL Note: Significant point AVILO at position: S265854.69 E0284315.62 (R170/56 DME JSV)	Leave AVILO on R170 JSV (INBD). At 29 DME JSV on R170 JSV turn left to intercept R310 STV (OUBD). Crossing R045 HGV turn left to intercept R030 HGV (INBD) for radar vectoring onto the ILS LOC RWY 03R. Pass AVILO 250KT IAS or less. Pass 30 DME JSV 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing the Outer Marker at 150 KT IAS, or as cleared by ATC.

Clearance	Details
Communication Failure (AVILO 1A)(Squawk 7600)	
Before AVILO Proceed to STV and enter the STV hold descending to FL240, or if lower than FL240, last assigned level and hold for minimum 5 MIN. Leave STV on the "After STV" Communication Failure Procedure. After AVILO Leave AVILO on R170 JSV (INBD) and descend to FL130. At R170/29 DME JSV turn left to intercept R310 STV (OUBD) and descend to FL110. On crossing R030 HGV descend to 8000' ALT. Crossing R197 JSV turn right onto track 350° to intercept the ILS LOC RWY 03R and complete a straight-in ILS approach and land RWY 03R. Note Aircraft to route to the STV hold in Communication Failure Procedure. Caution Holding patterns below FL110 will be conducted partially outside of controlled airspace.	
AVAGO ONE CHARLIE RNAV (GNSS) ARRIVAL (AVAGO 1C)	
AVAGO 1C RNAV (GNSS) ARRIVAL Note: Significant point AVAGO at position: S254309.90 E0272639.36 (R320/50 DME JSV)	Leave AVAGO for JS035. At JS035 turn right for JS036. At JS036 turn right proceeding to JS037 for radar vectoring onto the ILS LOC RWY 03R. Pass AVAGO 250KT IAS or less. Pass JS035 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing the outer maker at 150KT IAS or as cleared by ATC.
Communication Failure (AVAGO 1C)(Squawk 7600)	
Before AVAGO Proceed to AVAGO and enter the AVAGO RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL130 in the hold, or maintain last assigned level if below FL130. Leave AVAGO on the "After AVAGO" Communication Failure Procedure. After AVAGO Continue on the AVAGO 1C RNAV (GNSS) STAR. At JS035 descend to FL090, at JS037 descend to 8000' ALT. At JS037 proceed to VEKOP and continue with the RNAV (GNSS) RWY 03R to intercept the ILS LOC RWY 03R and land RWY 03R. Note Aircraft entering the TMA at FL110 and below are to enter the AVAGO RNAV (GNSS) hold at last assigned level, and continue on the routing for the AVAGO 1C STAR. Caution Holding patterns below FL110 will be conducted partially outside of controlled airspace. Pilots to take note of the appropriate FAD's, FAP's, FAR's as well as the Magalies Glider Window (FL110-FL145).	

Clearance	Details
NIBEX ONE BRAVO RNAV (GNSS) ARRIVAL (NIBEX 1B)	
NIBEX 1B ARRIVAL Note: Significant point NIBEX at position: S264925.56 E02740'3.41 (R235/50DME JSV)	Leave NIBEX for JS012. At JS012 turn right to JS013. At JS013 turn left to JS014. At JS014 turn left proceeding to JS015 and intercept the ILS LOC RWY 03R.
Communication Failure (NIBEX 1B)(Squawk 7600)	
Before NIBEX Proceed to NIBEX and enter the NIBEX RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL130 in the hold, or maintain last assigned level if below FL130. Leave NIBEX on the "After NIBEX" Communication Failure Procedure. After NIBEX Continue on the NIBEX 1B RNAV (GNSS) STAR to JS012. At JS012 descend to FL100, at JS013 descend to FL090, at JS014 adjust to 9000' ALT. At JS015 complete a straight-in ILS approach and land RWY 03R. Note Aircraft entering the TMA at FL110 and below are to enter the NIBEX RNAV (GNSS) hold at last assigned level, and continue the routing for the NIBEX 1B STAR. Caution Holding patterns below FL110 will be conducted partially outside of controlled airspace.	
AVILO ONE BRAVO ARRIVAL (AVILO 1B)	
AVILO 1B ARRIVAL Note: Significant point AVILO at position: S265854.69 E0284315.62	Leave AVILO for JS018. At JS018 turn left for JS017. At JS017 turn left proceeding to ETLIG for radar vectoring onto the ILS LOC RWY 03.
Communication Failure (AVILO 1B)(Squawk 7600)	
Before AVILO Proceed to STV and enter the STANDERTON RNAV (GNSS) hold descending to FL240, or if lower than FL240, last assigned level and hold for minimum 5 MIN. Leave STV on the "After STV" Communication Failure Procedure. After AVILO Continue on the AVILO 1B RNAV (GNSS) STAR. At ETLIG continue on the RNAV (GNSS) RWY 03R to JS2F1 and intercept the ILS LOC RWY 03R and land RWY 03R. Note: Nil Caution: Nil	

6 JOHANNESBURG FIR

O R Tambo International Airport

All STAR applicable for runway 21L. Use STAR for runway 21R only when instructed by ATC or announced on ATIS.

6.1 Standard Terminal Arrival Routes (STAR) RWY 21L

6.1.1 Applicable to the following aircraft:

a) Aircraft already cleared for the STAR.
b) All aircraft inside the Johannesburg FIR with flight plan destinations as O R Tambo International, and able to comply with a STAR.
c) When Communication Failure is experienced after being cleared for the approach, continue with the approach and land on the designated RWY.
d) Speed limit points (SLP) will be defined per procedure. If the speed is below minimum safe operating speed, the minimum safe operating speed will be flown and ATC advised. Unless for emergency pilots are not to request cancellation of speed restrictions.

6.1.2 Communication Failure Procedures for STAR.

If not cleared for an arrival, proceed to the nearest STAR entry position at last assigned level or FL090, whichever is the highest. Comply with the Communication Failure procedure associated with that STAR position.

6.1.3 Provisions

a) Aircraft must be able to receive the ATIS.
b) SIDs and STARs are announced in operation on ATIS.
c) Runway 21L is in use on the ATIS.
d) SIDs and STARs will only be in force when surveillance radar in operation.
e) If unable to comply with SID or STAR, notify ATC.
f) General Aviation Traffic operates below the TMA up to 7500FT ALT and is considered separated from traffic operating in the TMA.

RUNWAY 21L

Clearance	Details
AVAGO ONE DELTA RNAV (GNSS) ARRIVAL (AVAGO 1D)	
AVAGO 1D ARRIVAL Note: Significant point AVAGO at position: 254309.90S 0272639.36E (R320/50DME JSV)	To OR364 at FL110 or above, to OR365 at 8000FT or above, turn left to OR366 at 8000FT or above. At OR366 continue on track 035°M, expect radar vectors RWY 21L.
Communication Failure (AVAGO 1D)(Squawk 7600)	
Before AVAGO Proceed to AVAGO and enter the AVAGO 1D RNAV (GNSS) hold. Hold for 5 minutes at last assigned level (at or above FL110) Descend to FL130 in the hold or maintain last assigned level if below FL130. Leave AVAGO on the "After AVAGO" Communication Failure Procedure. After AVAGO Continue on the AVAGO 1D RNAV (GNSS) STAR. At OR366 to UVLOG. At UVLOG, continue on the RNAV-02 APCH RWY 21L. Note NIL Caution NIL	

Clearance	Details
NIBEX ONE DELTA RNAV (GNSS) ARRIVAL (NIBEX 1D)	
Clearance	**Details**
NIBEX 1D ARRIVAL Note: Significant point NIBEX at position: 264925.56S 0274013.41E (R235/50DME JSV)	Leave NIBEX for JS023. At JS023 turn left proceeding to UVLOG for radar vectoring onto the ILS LOC RWY 21L. Pass NIBEX 250KT IAS or less. Pass JS023 210KT IAS or less. On LOC maintain 180KT IAS until 10 DME. Passing the Outer Marker at 150KT IAS, or as cleared by ATC.
Communication Failure (NIBEX 1D)(Squawk 7600)	
Before NIBEX Proceed to NIBEX and enter the NIBEX RNAV (GNSS) hold. Hold at last assigned level for minimum 5 MIN, then descend to FL130 in the hold, or maintain last assigned level if below FL130. Leave NIBEX on the "After NIBEX" Communication Failure Procedure. After NIBEX Continue on the NIBEX 1D RNAV (GNSS) STAR. At JS023 descend to FL090, at UVLOG descend to 8000' ALT. AT UVLOG continue on the RNAV (GNSS) RWY 21L to JS3F2 and intercept the ILS RWY 21L and land RWY 21L. Note: Aircraft entering the TMA at FL110 and below are to enter the NIBEX RNAV (GNSS) hold at last assigned level and continue on the routing for the NIBEX 1D STAR. Caution: Holding patterns below FL110 will be conducted partially outside of controlled airspace.	

7 DEPARTING FLIGHTS

GENERAL

IFR flights departing from controlled aerodromes will receive initial ATC clearance from the local aerodrome control tower. The clearance limit will normally be the aerodrome of destination.

IFR flights departing from other aerodromes can take off without prior arrangements with the Area Control Centre concerned, unless controlled airspace is to be entered immediately after take-off. In such cases arrangements must be made to obtain a clearance before the controlled airspace is entered.

Aircraft noise during take-off and landing creates an ever increasing annoyance to inhabitants of areas adjacent to aerodromes. Continuous efforts are made to devise procedures in order to reduce aircraft noise without jeopardising safe operations

In order to reduce or to restrict aircraft noise to the minimum, the following procedures are proposed:-

– Turnouts immediately after take-off must as far as possible be avoided and runway heading must be maintained to a reasonable altitude.
– Where possible, aircraft must climb at the best angle of climb after take-off and this must be maintained until all built-up areas are over-flown, or the desired altitude has been reached/attained.

– Testing and run-ups of aircraft engines must as far as possible be avoided during the period 2000 and 0400.
– An appeal is made to all pilots to be considerate towards inhabitants of areas adjacent to aerodromes by combating aircraft noise at all times.

7.1 NOISE ABATEMENT PROCEDUREs

Two examples of operating procedures for the climb have been developed by ICAO as guidance. These two examples of Noise Abatement procedures can be defined as follows:

NADP 1 - The first example (NADP 1) is intended to describe one method, but not the only method, of providing noise reduction for noise-sensitive areas in close proximity to the departure end of the runway.

NADP 2 - The second example (NADP 2) similarly describes one method but not the only method, of providing noise reduction to areas more distant from the runway end.

These procedures applies to jet aircraft and may be disregarded if:

At 3000FT AGL or when levelled off by ATC, or when levelled by the Standard Instrument Departure (SID)

The following procedure will be in force at the listed airport:

King Shaka International Airport (Durban)	NADP 1
Lanseria International Airport	NADP 1

NADP 1

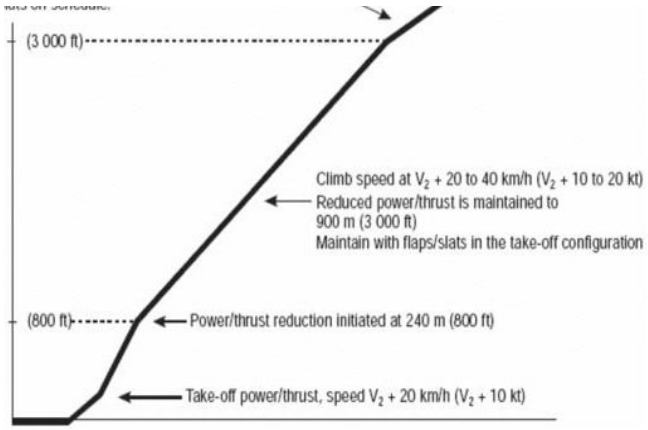
1) TAKE-OFF TO 800FT AGL
   i) Take-off power/thrust
   ii) Speed, climb at V2+10knots (V2+20km/h)

   Note: Dependant on aircraft type the take-off power/thrust may be reduced at a lower height.

2) AT 800FT AGL
   i) Power/thrust reduction is initiated in accordance with the noise abatement power/ thrust schedule provided in the aircraft operating manual.

3) 800FT AGL TO 3000FT AGL
   i) Climb at V2+10 to 20km/h).
   ii) Reduced power/thrust is maintained to 900M (3000FT)
   iii) Maintain with flaps/slats in take-off configuration.

4) AT 3000FT AGL
   i) On reaching 900M (3000FT) maintain positive rate of climb.
   ii) Accelerate smoothly to en-route climb speed
   iii) Retract flaps/slats on schedule.
   iv) In addition, all aircraft at or below FL100 will fly a maximum speed of 250KIAS. No jet aircraft are to use RWY or TWY intersection for take-off at any airport within South Africa between the hours of 2000-0400.



Noise abatement take-off climb - Example of a procedure alleviating noise close to the departure (NADP1)

The following procedures will be in force at the listed airports:

Bram Fischer International Airport. (Bloemfontein)	NADP 2
Cape Town International Airport.	NADP 2
King Phalo Airport (East London)	NADP 2
George Airport	NADP 2

Kimberley Airport	NADP 2
O R Tambo International Airport	NADP 2
Chief Dawid Stuurman International Airport (Port Elizabeth)	NADP 2
Upington International Airport	NADP 2
Wonderboom Airport (Pretoria)	NADP 2

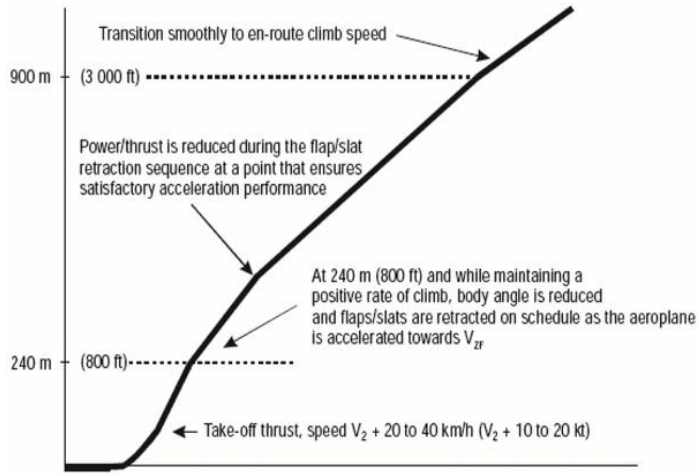
NADP 2

1) Take-off to 800FT AGL
   i) Take-off power/thrust
   ii) Speed, climb at V2+10 to 20knots (V2+20 to 40km/h)

   Note: Dependant on aircraft type, the take-off power/thrust may be reduced at a lower height.

2) At 800FT AGL
   i) While maintaining a positive rate of climb, body angle reduced.
   ii) Flaps/Slats are retracted on schedule
   iii) Aircraft is accelerated towards VZF.

3) 800FT AGL to 3000FT AGL
   i) Power/Thrust is reduced during the flap/slat retraction sequence at a point that ensures satisfactory acceleration performance.
   ii) At 3000FT AGL
   iii) On reaching 900M (3000FT), transition smoothly to en-route climb speed.



Noise abatement take-off climb - Example of a procedure alleviating noise distant from the aerodrome (NADP 2)

7.2 NOISE ABATEMENT PROCEDURES - WATERKLOOF AIR FORCE BASE (MIL UNLICENSED)

1) GENERAL
The following procedures are designed to avoid excessive noise in areas adjacent to the aerodrome and in areas overflown during take-off and landing. Strict adherence within the limits of safety and performance is required. Aircraft not landing at Waterkloof are not permitted to overfly built up areas below 1000 FT AGL.

a) Rapid changes in engine power should be avoided.
b) Heavy category aircraft commanders are permitted to select the runway for arrivals and departures.
c) Bad weather circuits (500FT) may be flown in IMC or two per training/rating sortie.
d) Heavy category aircraft are not permitted to practice 500FT circuits.

2) ARRIVALS
   a) All IFR arrivals, wind and traffic permitting, will be executed on RWY01.
   b) The final approach path shall be strictly at the angle defined by the ILS and PAPI glide path (3°).
   c) VFR flights must adhere to the glide path on final approach and where possible avoid built up areas.
   d) Fighter aircraft flying via initial point may not exceed 400KTS IAS.

3) DEPARTURES
   a) VFR flights will climb to 1000FT AGL as soon as possible and once clear of built up areas ascend accordingly.

4) TAKE-OFF TO 1500 FT AGL
   a) Take-off power.
   b) Take-off flaps.
   c) Climb at V2 + 10 to 20KTS or as limited by body angle. Dependent on ACFT type, take-off power/thrust may be reduced at a lower altitude.

5) AT 1500 FT AGL
   a) Reduce thrust (if not reduced already) to not less than climb power/thrust.

6) 1500 FT AGL TO 3000 FT AGL
   a) Climb at V2 + 10 to 20KTS.

7) AT 3000 FT AGL
   a) Accelerate smoothly to en-route climb speed with flap retraction on schedule.

8) ENGINE RUN-UP TESTS
   Engine run-up tests on Saturdays, Sundays and public holidays and on working days between 1600 and 0400 will only be permitted with the authorization of the operations coordinator.

9) EXEMPTIONS

Restrictions to arrivals/departures do not apply in the following cases.

a) In an emergency or simulated emergency.
b) For operational flights.
c) For inspection and calibration flights.
d) For flights with the prior authorization from the Air Wing Coordinator AFB Waterkloof.

10) NIGHT FLYING TRAINING - MONDAY TO THURSDAY

Night flying training must be authorised by the Air Wing Coordinator.

a) Final arrival: 2000.
b) No circuit and landings between 1900 and 2000.

11) FLYING TRAINING - SATURDAYS, SUNDAYS AND PUBLIC HOLIDAYS

i) Saturdays: Flying training authorised by the Air Wing Coordinator may take place between 0600 and 1600.
ii) Sundays: No flying training may take place except with prior authorization by CAF.
iii) Public Holidays: No flying training may take place.

8 CAPE TOWN INTERNATIONAL AIRPORT

8.1 Standard Instrument Departures (SID) RWY01

8.1.1 Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in paragraph 8.1. In addition, all aircraft at or below FL 100 will fly a MAXIMUM speed of 250 KIAS.

8.1.2 At 2000 FT ALT contact Cape Town Radar (APP) on the frequency provided in the ATC clearance. Advise RADAR (APP) of level passing on first contact for Mode-C check.

8.1.3 Cross CTR boundary at or above 2500 FT ALT.

8.1.4 Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.

8.1.5 If unable to comply with SID notify ATC.

8.1.6 Communication Failure Procedure for SID

a) Except in the event of an emergency, aircraft shall comply with the communication failure procedures for each SID irrespective of VMC or IMC.
b) Comply with the SID procedure, climb to MSA or maintain last assigned flight level, whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 01

Clearance	RWY	Turn	Details
KODES 1A DEPARTURE (KODES 1A)			
KODES 1A DEPARTURE (KODES 1A) NOTE: Significant point KODES at position: S330840 E0184520 (R-032/50DME "CTV")	01	Right	Climb to FL090, maintain RWY track to 3DME "CTV" then turn left to track 330°M to intercept R-354 "CTV" (outbound). Passing 8DME "CTV" turn right to track 060°M to intercept R-032 "CTV" (outbound) direct KODES. Further climb will be under radar control. At KODES set course as per flight plan.
Communication Failure (KODES 1A) (Squawk 7600)			
Comply with the KODES 1A SID, maintain last assigned level. Passing 30DME "CTV" climb to flight plan level. At KODES continue as per flight plan. Aircraft wishing to return must continue to the SID termination point at FL090, then proceed to ERDAS and comply with the appropriate ERDAS STAR Communication Failure procedure.			
IMSOM 1A DEPARTURE (IMSOM 1A)			
IMSOM 1A DEPARTURE (IMSOM 1A) NOTE: Significant point IMSOM at position:S335603.8989 E0180031.2411(R-283/30DME "CTV")	01	Left	Climb to FL090, maintain RWY track to 3DME "CTV" then turn left to track 330°M to intercept R-354 "CTV" (outbound). Passing 8DME "CTV" turn left direct IMSOM. Further climb will be under radar control. At IMSOM set course as per flight plan.
Communication Failure (IMSOM 1A) (Squawk 7600)			
Comply with the IMSOM 1A SID, maintain last assigned level. At IMSOM continue as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point at FL090, then proceed to "ASPIK" and comply with the appropriate "ASPIK" STAR Communication Failure procedure			

8.2 Standard Instrument Departures (SID) RWY19.

8.2.1 Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in paragraph 8.1. In addition, all aircraft at or below FL 100 will fly a MAXIMUM speed of 250 KIAS.

8.2.2 At 2000 FT ALT contact Cape Town Radar (APP) on the frequency provided in the ATC clearance. Advise RADAR (APP) of level passing on first contact for Mode-C check.

8.2.3 Cross CTR boundary at or above 2500 FT ALT.

8.2.4 Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.

8.2.5 If unable to comply with SID notify ATC.

8.2.6 Communication Failure Procedure for SID

a) Except in the event of an emergency, aircraft shall comply with the communication failure procedures for each SID irrespective of VMC or IMC.

b) Comply with the SID procedure, climb to MSA or maintain last assigned flight level, whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 19

Clearance	RWY	Turn	Details
TETAN 1A DEPARTURE (TETAN 1A)			
TETAN 1A DEPARTURE (TETAN 1A) NOTE: Significant point TETAN at NDB "WC" position: S333948.00 E0192458.00 (R-089/44.5DME "CTV") 6.2%@ 140KTAS= 880ft/min 6.2%@ 180KTAS= 1130ft/min 6.2%@ 220KTAS= 1381ft/min	19	Left	Restricted to a minimum climb gradient of 6.2% until passing FL085 due to terrain. Do not cross (North of) "CTV" R-140 below FL085. Climb to FL090, maintain RWY track to 5DME "CTV" then turn left to track 145°M. Passing 18DME "CTV" turn left direct TETAN. Further climb will be under radar control. At TETAN set course as per flight plan.
Communication Failure (TETAN 1A) (Squawk 7600)			
Comply with the TETAN 1A SID, maintain last assigned level. Crossing R-105 "CTV" climb to flight plan level. At TETAN continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and then route direct ERDAS and comply with the appropriate ERDAS STAR Communication Failure procedure.			
TETAN 1B DEPARTURE (TETAN 1B)			
TETAN 1B DEPARTURE (TETAN 1B) NOTE: Significant point TETAN at NDB "WC" position: S333948.00E0192458.00 (R-089/44.5DME "CTV") 4.6%@ 140KTAS=652ft/min 4.6%@ 180KTAS= 838ft/min 4.6%@ 220KTAS=1024ft/min	19	Left	Restricted to a minimum climb gradient of 4.6% until passing 5700 FT ALT due to terrain. Do not cross (north of) R-140 below 8500 FT ALT. Climb to FL090, maintain RWY track to 5DME then adjust onto R-190 "CTV" (outbound). Passing 20DME "CTV" turn left to track 100°M. Crossing R-170 "CTV" turn left direct TETAN. Further climb will be under radar control. At TETAN set course as per flight plan.
Communication Failure (TETAN 1B) (Squawk 7600)			
Comply with the TETAN 1B DEPARTURE and maintain last assigned level. Crossing R-105 "CTV" climb to flight plan level. At TETAN continue as per flight plan.Aircraft wishing to return must continue to the SID termination point and then route direct ERDAS and comply with the appropriate ERDAS STAR Communication Failure procedure.			

Clearance	RWY	Turn	Details
IMSOM 1B DEPARTURE (IMSOM 1B)			
IMSOM 1B DEPARTURE (IMSOM 1B) NOTE: Significant point IMSOM at position: S335603.8989 E0180031.2411 (R-283/30DME "CTV")	19	Right	Climb to FL090, maintain RWY track to 5DME "CTV" then turn right to track 240°M to intercept R-210 "CTV" (outbound). Passing 5500 FT ALT turn right direct ASPIK, then direct IMSOM. Further climb will be under radar control. At IMSOM set course as per flight plan.
Communication Failure (IMSOM 1B) (Squawk 7600)			
Comply with the IMSOM 1B SID, maintain last assigned level. At IMSOM continue as per flight plan and climb to flight plan level.Aircraft wishing to return must continue to the SID termination point at FL090, then proceed to ASPIK and comply with the appropriate ASPIK STAR Communication Failure procedure.			
OKTED 1B DEPARTURE (OKTED 1B)			
OKTED 1B DEPARTURE (OKTED 1B) NOTE: Significant point OKTED at Position: S341312.000 E0193330.000 (R-131/50 DME "CTV") 6.2%@ 140KTAS= 880ft/min 6.2%@ 180KTAS= 1130ft/min 6.2%@ 220KTAS=1381ft/min	19	Left	Restricted to a minimum climb gradient of 6.2% until passing FL085 due to terrain. Do not cross (North of) "CTV" R-140 below FL085. Climb to FL090, maintain RWY track to 5DME "CTV" then turn left to track 145°M. Passing 18DME "CTV" turn left direct OKTED. Further climb will be under radar control. At OKTED continue as per flight plan.
Communication Failure (OKTED 1B) (Squawk 7600)			
Comply with the OKTED 1B SID, maintain last assigned level. At OKTED continue as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and then route direct GETEN and comply with the appropriate STAR Communication Failure procedure.			
OKTED 1C DEPARTURE (OKTED 1C)			
OKTED 1C DEPARTURE (OKTED 1C) NOTE: Significant point OKTED at position: S341312.000 E0193330.000 (R-131/50 DME "CTV") 4.6%@ 140KTAS= 652ft/min 4.6%@ 180KTAS= 838ft/min 4.6%@ 220KTAS=1024ft/min	19	Left	Restricted to a minimum climb gradient of 4.6% until passing 5700 FT ALT due to terrain. Do not cross (north of)"CTV" R-140 below FL085. Climb to FL090, maintain RWY track to 5DME then adjust onto R-190 "CTV" (outbound). Passing 20DME "CTV" turn left to track 100°M. Crossing R-170 "CTV" turn left direct OKTED. Further climb will be under radar control. At OKTED set course as per flight plan.
Communication Failure (OKTED 1C) (Squawk 7600)			
Comply with the OKTED 1C SID, maintain last assigned level. At OKTED continue as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and then route direct GETEN and comply with the appropriate STAR Communication Failure procedure.			

9 JOHANNESBURG FIR

King Shaka International Airport

9.1 STANDARD INSTRUMENT DEPARTURES (SID) RWY 06.

a) If unable to comply with SID notify ATC.

b) Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in ENR 1.5 paragraph 8.1

c) At 2000 FT ALT contact Durban Radar (APP) on the frequency provided in the ATC clearance. Advise Radar (APP) of level passing on first contact for Mode-C check.

d) Cross CTR BDRY at or above 3000 FT ALT.

e) Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.

9.2 COMMUNICATION FAILURE PROCEDURE FOR SID:

a) Except in the event of an emergency, aircraft shall comply with the Communication Failure procedures for each SID irrespective of VMC or IMC.

b) Comply with the SID procedure, climb to MSA or maintain last assigned flight level whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 06

Clearance	Details
ITMIL ONE CHARLIE RNAV (GNSS) DEPARTURE (ITMIL 1C)	
ITMIL 1C RNAV (GNSS) DEPARTURE Note: Significant point ITMIL at position: S290151.05 E0314838.40 (R070/50 DME TGV) 7.0%ROC @140KT IAS = 992 FPM 7.0%ROC @180KT IAS = 1276 FPM 7.0%ROC @220KT IAS = 1560 FPM	Climb to FL070. Maintain RWY track to LE015. At LE015 turn right to ITMIL. Further climb will be under radar control. At ITMIL set course as per flight plan.
Communication Failure (ITMIL 1C) (Squawk 7600)	
Maintain RWY track LE015. At LE015 turn right to ITMIL maintaining last assigned level or MSA whichever is higher. Passing LE017 climb to flight plan level. At ITMIL set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level or MSA whichever is higher. Enter the ITMIL hold and hold for 5 MIN. Comply with the ITMIL RNAV (GNSS) STAR Communication Failure Procedure.	

Clearance	Details
TUBIN ONE ALPHA RNAV (GNSS) DEPARTURE (TUBIN 1A)	
TUBIN 1A RNAV (GNSS) DEPARTURE Note: Significant point TUBIN at position: S300201.29 E0301754.66 (R262/50 DME TGV) 7.0%ROC @140KT IAS = 992 FPM 7.0%ROC @180KT IAS = 1276 FPM 7.0%ROC @220KT IAS = 1560 FPM	Climb to FL070. Maintain RWY track to LE015. At LE015 turn left direct to LE1G2. At LE1G2 turn left onto track and proceed to LE1CF. At LE1CF proceed to TUBIN. Further climb will be under radar control. At TUBIN set course as per flight plan.
Communication Failure (TUBIN 1A) (Squawk 7600)	
Maintain RWY track to LE015. At LE015 turn left direct to LE1G2. At LE1G2 turn left to LE1CF. At LE1CF proceed to TUBIN maintaining last assigned flight level or MSA whichever is higher. Passing LE1CF climb to flight plan level. At TUBIN set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level or MSA or FL080 whichever is higher. At TUBIN proceed to DUNSA and comply with the DUNSA 1A STAR Communication Failure Procedure.	
GREYTOWN ONE ALPHA RNAV (GNSS) DEPARTURE (GYV 1A)	
GREYTOWN 1A RNAV (GNSS) DEPARTURE Note: Significant point GYV at position: S290728.99 E0303508.06 (R340/41 DME TGV) 7% @140KT IAS = 992 FPM 7% @180KT IAS = 1276 FPM 7% @220KT IAS = 1560 FPM	Climb to FL070. Maintain RWY track to LE015. At LE015 turn left direct to LE3CF. At LE3CF proceed to GYV. Further climb will be under radar control. At GYV set course as per flight plan.
Communication Failure (GYV 1A) (Squawk 7600)	
Maintain RWY track to LE015. At LE015 turn left direct to LE3CF maintaining last assigned flight level or MSA whichever is higher. Passing LE3CF climb to flight plan level and proceed to GYV. At GYV set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level or MSA or FL080 whichever is higher. At GYV proceed to GETOK and comply with the GETOK 1C STAR Communication Failure Procedure.	

King Shaka International Airport

1) STANDARD INSTRUMENT DEPARTURES (SID) RWY 24.
   a) If unable to comply with SID notify ATC.
   b) Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in ENR 1.5 paragraph 8.1
   c) At 2000 FT ALT contact Durban Radar (APP) on the frequency provided in the ATC clearance. Advise Radar (APP) of level passing on first contact for Mode-C check.
   d) Cross CTR BDRY at or above 3000 FT ALT.
   e) Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.
2) COMMUNICATION FAILURE PROCEDURE FOR SID:
   a) Except in the event of an emergency, aircraft shall comply with the Communication Failure procedures for each SID irrespective of VMC or IMC.
   b) Comply with the SID procedure, climb to MSA or maintain last assigned flight level whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 24

Clearance	Details
OKTAN ONE ALPHA RNAV (GNSS) DEPARTURE (OKTAN 1A)	
OKTAN 1A RNAV (GNSS) DEPARTURE Note: Significant point OKTAN at position: S301758.32 E0303453.34 (R239/50 DME TGV) 7.0%ROC @140KT IAS = 992 FPM 7.0%ROC @180KT IAS = 1276 FPM 7.0%ROC @220KT IAS = 1560 FPM	Climb to FL070. Proceed direct to OKTAN. Further climb will be under radar control. At OKTAN set course as per flight plan.
Communication Failure (OKTAN 1A) (Squawk 7600)	
Maintain RWY track to LE018 maintaining last assigned flight level or MSA, whichever is higher. At LE018 proceed to OKTAN and climb to flight plan level. At OKTAN set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level, MSA or FL080 whichever is higher. Enter the OKTAN hold and hold for 5 MIN. Proceed to TGV and comply with the ILS Y RWY 24 Procedure.	
TUBIN ONE BRAVO RNAV (GNSS) DEPARTURE (TUBIN 1B)	
TUBIN 1B RNAV (GNSS) DEPARTURE Note: Significant point TUBIN at position: S300201.29 E0301754.66 (R262/50 DME TGV) 7% @140KT IAS = 992 FPM 7% @180KT IAS = 1276 FPM 7% @220KT IAS = 1560 FPM	Climb to FL070. Maintain RWY track to LE2D2. At LE2D2 turn right direct to LE2CF. At LE2CF proceed to TUBIN. Further climb will be under radar control. At TUBIN set course as per flight plan.

Clearance	Details
Communication Failure (TUBIN 1B) (Squawk 7600)	
Maintain RWY track to LE2D2 maintaining last assigned flight level or MSA, whichever is higher. At LE2D2 turn right direct to LE2CF and climb to flight plan level. At TUBIN set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level, MSA or FL080 whichever is higher. At TUBIN proceed to DUNSA and comply with the DUNSA 1B STAR Communication Failure Procedure.	
GREYTOWN TWO BRAVO RNAV (GNSS) DEPARTURE (GYV 2B)	
GREYTOWN 2B RNAV (GNSS) DEPARTURE Note: Significant point GYV at position: S290728.99 E0303508.06 (R340/41 DME TGV) 7% @140KT IAS = 992 FPM 7% @180KT IAS = 1276 FPM 7% @220KT IAS = 1560 FPM	Climb to FL070. Maintain RWY track to LE2D2. At LE2D2 turn right direct to LE4CF. At LE4CF proceed to GYV. Further climb will be under radar control. At GYV set course as per flight plan.
Communication Failure (GYV 2B) (Squawk 7600)	
Maintain RWY track to LE2D2. At LE2D2 turn right to LE4CF maintaining last flight level or MSA whichever is higher. Passing LE4CF climb to flight plan level and proceed to GYV. At GYV set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to the last assigned level or MSA or FL080 whichever is higher. At GYV proceed to GETOK and comply with the GETOK 1D STAR Communication Failure Procedure.	

10 JOHANNESBURG FIR

O R TAMBO INTERNATIONAL AIRPORTNOTE: SIMULTANEOUS USE OF PARALLEL RUNWAYS 03L/ 03R AND 21L/21R - O R TAMBO INTERNATIONAL AIRPORT

10.1 STANDARD INSTRUMENT DEPARTURES (SID) RWY 03.

10.1.1 Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in ENR 1.5 paragraph 8.1

10.1.2 At 6 500FT ALT contact Johannesburg Radar (APP) on the frequency provided in the ATC clearance. Advise Radar (APP) of level passing on first contact for Mode-C check.

10.1.3 Cross CTR BDRY at or above 8 000 FT ALT.

10.1.4 Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.

10.1.5 If unable to comply with SID notify ATC.

10.2 COMMUNICATION FAILURE PROCEDURE FOR SID:

10.2.1 Except in the event of an emergency, aircraft shall comply with the communication failure procedures for each SID irrespective of VMC or IMC.

10.2.2 Comply with the SID procedure, climb to MSA or maintain last assigned flight level, whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 03L

Clearance	RWY	Turn	Details
EGMEN TWO ALPHA DEPARTURE (EGMEN 2A)			
EGMEN 2A DEPARTURE Note: Significant point EGMEN at position: S253845.74 E0285805.53 (R070/50 DME JSV) 4.5% @140KT IAS = 638 FPM 4.5% @180KT IAS = 820 FPM 4.5% @220KT IAS =1003 FPM	03L	Right	Restricted to a minimum climb gradient of 4.5% to the CTR boundary. Climb to FL090, maintain RWY track to 5.5 DME JSV then turn right to intercept R275 WIV (INBD). At 23.4 DME JSV turn left onto track 030° to intercept R070 JSV (OUBD) to EGMEN. Cross 15 DME JSV at FL090 or above.
Communication Failure (EGMEN 2A) (Squawk 7600)			
Comply with EGMEN 2A SID, climbing to FL090 or maintain last assigned level, whichever is highest. At EGMEN set course as per flight plan and climb to flight plan level Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if cleared level is below MSA. At EGMEN proceed to OKPIT and comply with the OKPIT 4A RWY03R STAR Communication Failure Procedure.			
APDAK THREE ALPHA DEPARTURE (APDAK 3A)			
APDAK 3A DEPARTURE Note: Significant point APDAK at position: S263303.55 E0290252.64 (R152/62 DME WKV) (R137/50 DME JSV) 4.5% @140 KT IAS = 638 FPM 4.5% @180 KT IAS = 820 FPM 4.5% @220 KT IAS =1003 FPM	03L	Right	Restricted to a minimum climb gradient of 4.5% to the CTR boundary. Climb to FL090, maintain RWY track to 5.5 DME JSV then turn right onto track 101° to intercept R152 WKV (OUBD) to APDAK. Cross 15 DME JSV at FL090 or above.
Communication Failure (APDAK 3A) (Squawk 7600)			
Comply with the APDAK 3A SID, climbing to FL090 or maintain last assigned level, whichever is the highest. At APDAK continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At APDAK proceed to STV and comply with the STANDERTON 6A RWY03R STAR Communication Failure Procedure.			
VASUR THREE ALPHA DEPARTURE (VASUR 3A)			
VASUR 3A DEPARTURE Note: Significant point VASUR at position: 253346.75S 0275335.87E (R351/40DME JSV) 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	03L	Right	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 5.5 DME JSV then turn left onto track 311° to intercept R351 JSV (OUBD) to VASUR. Further climb will be under radar control. At VASUR set course as per flight plan.

Clearance	RWY	Turn	Details
Communication Failure (VASUR 3A) (Squawk 7600)			
Comply with VASUR 3A SID, climbing to 8700' ALT or maintain last assigned level, whichever is highest. At VASUR set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At VASUR proceed to AVAGO and comply with the AVAGO 2A RWY03 STAR Communication Failure Procedure.			
RAGUL THREE ALPHA DEPARTURE (RAGUL 3A)			
RAGUL 3A DEPARTURE Note: Significant point RAGUL at position: S262654.35 E0273443.89 (R241/50DME WKV) 4.4% @140KT IAS = 624 FPM 4.4% @180KT IAS = 802 FPM 4.4% @220KT IAS = 980 FPM	03L/R	Left	Restricted to a minimum climb gradient of 4.4% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 5.5 DME JSV. At 5.5 DME JSV turn left onto track 281° to intercept R241 WKV (OUBD) to RAGUL. Further climb will be under RADAR control. At RAGUL set course as per flight plan.
Communication Failure (RAGUL 3A) (Squawk 7600)			
Comply with the RAGUL THREE ALFA SID, climbing to 8700' ALT or maintain last assigned level, whichever is highest. At RAGUL continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At RAGUL proceed to NIBEX and comply with the NIBEX 2A RWY03R STAR Communication Failure Procedure.			
NESAN ONE ALFA DEPARTURE (NESAN 1A)			
NESAN 1A DEPARTURE Note: Significant point NESAN at position: S252948.88 E0280705.56 (R010/40DME JSV) 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	03L/R	Left	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 8 DME JSV then turn left direct to NESAN. Further climb will be under radar control. At NESAN set course as per flight plan.
Communication Failure (NESAN 1A) (Squawk 7600			
Comply with the NESAN ONE ALPHA SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At NESAN continue as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At NESAN proceed to OKPIT and comply with the OKPIT 4A RWY03R STAR Communication Failure Procedure.			

Clearance	RWY	Turn	Details
GRASMERE FIVE BRAVO DEPARTURE (GRASMERE 5B) RESTRICTED TO TURBO PROP AIRCRAFT ONLY.			
GRASMERE 5B DEPARTURE Note: Significant point GAV at position:S263053. 63 E0274035.17 5.0% @ 80KT IAS = 405 FPM 5.0% @120KT IAS = 608 FPM 5.0% @160KT IAS = 810FPM 5.0% @200KT IAS =1013FPM	03L	Left	Restricted to a minimum climb gradient of 5.0% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 3 DME JSV or 6100' ALT, whichever is later, then turn left onto track 261°. Crossing 13 DME JSV turn left direct to GAV. Further climb will be under radar control. At GAV set course as per flight plan.
Communication Failure (GRASMERE 5B) (Squawk 7600)			
Comply with the GRASMERE FIVE BRAVO SID, climbing to 8700' ALT or maintain last assigned level, whichever is highest. At GAV set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At GAV proceed to NIBEX and comply with the NIBEX 2A RWY03R STAR Communication Failure Procedure.			
EXOBI ONE ALFA DEPARTURE (EXOBI 1A)			
EXOBI 1A DEPARTURE Note: Significant point "EXOBI" at position: S260252.2666 E0290851.3211 (R-101/50 DME "JSV") 4.2% ROC @140KTAS= 595ft/min 4.2% ROC @180KTAS= 766ft/min 4.2% ROC @220KTAS= 936ft/min	03L/R	Right	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to FL090, maintain RWY track to 5.5 DME JSV then turn right onto track 101°. At 15 DME JSV turn right onto track 126° to intercept R101 JSV (OUBD) to EXOBI. Cross 15 DME JSV at FL090 or above if cleared by ATC. Further climb will be under radar control. At EXOBI set course as per flight plan.
Communication Failure (EXOBI 1A) (Squawk 7600			
Comply with the EXOBI ONE ALPHA SID, climbing to FL090 or maintain last assigned level, whichever is highest. At EXOBI set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At EXOBI proceed to WIV and comply with the WITBANK 4A RWY03R STAR Communication Failure Procedure.			
EGMEN ONE CHARLIE RNAV (GNSS) DEPARTURE (EGMEN 1C)			
EGMEN 1C DEPARTURE Note: Significant point EGMEN at position: S253845.74 E0285805.53 (R070/50DME JSV) 4.1% @140KT IAS = 581 FPM 4.1% @180KT IAS = 747 FPM 4.1% @220KT IAS = 913 FPM	03L	Left	Restricted to a minimum climb gradient of 4.1% to the CTR boundary. Climb to FL090, maintain RWY track to JS031. At JS031 turn right to JS032. At JS032 turn left direct to EGMEN. Further climb will be under radar control. At EGMEN set course as per flight plan.

Clearance	RWY	Turn	Details
Communication Failure (EGMEN 1C) (Squawk 7600)			
Comply with the EGMEN ONE CHARLIE SID, climbing to FL090 or maintain last assigned level, whichever is highest. At EGMEN set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At EGMEN proceed to OKPIT and comply with the OKPIT 4A RWY03R STAR Communication Failure Procedure.			
APDAK ONE BRAVO RNAV (GNSS) DEPARTURE (APDAK 1B)			
APDAK 1B DEPARTURE Note: Significant point APDAK at position: 263303.55S 0290252.64E (R137/50DME JSV) 4.2% @ 140KT IAS = 596 FPM 4.2% @ 180KT IAS = 766 FPM 4.2% @ 220KT IAS = 936 FPM Note: This procedure is CO-DEPENDENT on the UNPOM 1A RNAV(GNSS) STAR RWY 03R procedure which needs to be published in order for the APDAK 1B RNAV (GNSS) SID RWY 03L Communication Failure Procedure to be valid.	03L	Right	Restricted to a minimum climb gradient of 4.2%, until reaching CTR BDRY. Climb to FL110. To OR378 on course 035˚M, turn right direct to OR379 at or below FL090, to OR415 at or above FL110, to APDAK at or above FL110. At APDAK, set course as per flight plan.
Communication Failure (APDAK 1B) (Squawk 7600)			
Comply with the APDAK 1B RNAV (GNSS) SID RWY 03L. At APDAK set course as per flight plan and climb to flight plan level. Aircraft wishing to return to FAOR must continue as per the APDAK 1B RNAV (GNSS) SID to APDAK. At APDAK proceed to UNPOM and enter the UNPOM 1A RNAV (GNSS) hold at FL110. Hold for 5 minutes. Thereafter, comply with the After UNPOM 1A RNAV (GNSS) STAR Communication Failure Procedure.			

11 JOHANNESBURG FIR

O R TAMBO INTERNATIONAL AIRPORT NOTE: SIMULTANEOUS USE OF PARALLEL RUNWAYS 03L/ 03R AND 21L/21R - O R TAMBO INTERNATIONAL AIRPORT

11.1 STANDARD INSTRUMENT DEPARTURES (SID) RWY 21R

11.1.1 Unless otherwise authorised by ATC, the maximum speeds applicable shall comply with noise abatement procedures as laid down in ENR 1.5 paragraph 8.1

11.1.2 At 6500 FT ALT contact Johannesburg Radar (APP) on the frequency provided in the ATC clearance. Advise Radar (APP) of level passing on first contact for Mode-C check.

11.1.3 Cross CTR BDRY at or above 8 000 FT ALT.

11.1.4 Where climb gradient required is greater than 3.3% see provisions contained within every specific procedure.

11.2 COMMUNICATION FAILURE PROCEDURE FOR SID:

11.2.1 Except in the event of an emergency, aircraft shall comply with the communication failure procedures for each SID irrespective of VMC or IMC.

11.2.2 Comply with the SID procedure, climb to MSA or maintain last assigned flight level, whichever is highest. At the SID termination position set course as per flight plan and climb to flight plan level.

RUNWAY 21

Clearance	RWY	Turn	Details
EGMEN TWO BRAVO DEPARTURE (EGMEN 2B)			
EGMEN 2B DEPARTURE Note: Significant point EGMEN at position: S253845.74 0285805.53E (R070/50 DME JSV) 4.3% @140KT IAS = 610 FPM 4.3% @180KT IAS = 784 FPM 4.3% @220KT IAS = 958 FPM	21R	Left	Restricted to a minimum climb gradient of 4.3% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 4.5 DME JSV then turn left onto track 111°. Intercept R253 WIV (INBD). Crossing R016 STV turn left onto track 054° to intercept R024 STV (OUBD) direct to EGMEN.
Communication Failure (EGMEN 2B) (Squawk 7600)			
Comply with EGMEN 2B SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At EGMEN set course as per flight plan and climb to flight plan level.Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At EGMEN proceed to OKPIT and comply with the OKPIT 4B RWY21L STAR Communication Failure Procedure.			
APDAK TWO BRAVO DEPARTURE (APDAK 2B)			
APDAK 2B DEPARTURE Note: Significant point APDAK at position: 263303.55S 0290252.64E (R137/50 DME JSV) 4.2% @140 KT IAS = 596 FPM 4.2% @180 KT IAS = 766 FPM 4.2% @220 KT IAS =936 FPM	21R	Left	Restricted to a minimum climb gradient of 4.2% to CTR boundary. Climb to 8000' ALT, maintain RWY track to 4.5 DME JSV then turn left direct to APDAK. Cross 4.5 DME JSV at 7000' ALT or above. Further climb will be under radar control. At APDAK set course as per flight plan.
Communication Failure (APDAK 2B) (Squawk 7600)			
Comply with the APDAK 2B SID, climbing to 8300' ALT or maintain last assigned level, whichever is the highest. At APDAK continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At APDAK proceed to STV and comply with the appropriate STAR Communication Failure Procedure.			

Clearance	RWY	Turn	Details
VASUR THREE BRAVO DEPARTURE (VASUR 3B)			
VASUR 3B DEPARTURE Note: Significant point VASUR at position: 253346.76S 0275335.87E (R015/23DME LIV) 4.2% @140KT IAS = 596 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	21L/R	Right	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 9 DME JSV then turn right onto track 291°M. Passing 15 DME JSV turn right onto track 351°M to intercept R198 LIV (INBD). At LIV intercept R015 LIV (OUBD) to VASUR. Further climb will be under radar control. At VASUR set course as per flight plan.
Communication Failure (VASUR 3B) (Squawk 7600)			
Comply with VASUR 3B SID, climbing to 8700' ALT or maintain last assigned level, whichever is highest. At VASUR set course as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At VASUR proceed to AVAGO and comply with the AVAGO 2B RWY21 STAR Communication Failure Procedure.			
RAGUL ONE DELTA DEPARTURE (RAGUL 1D)			
RAGUL 1D DEPARTURE Note: Significant point RAGUL at position: 262654.35S 0273443.89E (FL110) 3.3% @140KT IAS = 468 FPM 3.3% @180KT IAS = 602 FPM 3.3% @220KT IAS = 736 FPM **Note:** This procedure is CO-DEPENDENT on the NIBEX 1D RNAV (GNSS) STAR RWY 21L procedure which needs to be published in order for the RAGUL 1D RNAV (GNSS) SID RWY 21R Communication Failure Procedure to be valid.	21R	Right	Restricted to a minimum climb gradient of 3.3%, until reaching CTR boundary. Climb to FL090. To OR389 on course 215°M, turn right to OR398 at FL090 or below, turn left to RAGUL at FL110 or above, set course as per flight plan.
Communication Failure Procedure (Squawk 7600)			
Comply with the **RAGUL ONE DELTA RNAV (GNSS) SID RWY 21R.** At RAGUL set course as per flight plan and climb to flight plan level. Aircraft wishing to return to FAOR must continue as per the RAGUL 1D RNAV (GNSS) SID to RAGUL. At RAGUL proceed to NIBEX and enter the NIBEX 1D RNAV (GNSS) STAR RWY 21L hold at FL110. Hold for 5 minutes. Thereafter, comply with the "**After NIBEX 1D RNAV (GNSS) STAR"** Communication failure procedure.			
OVALA ONE BRAVO DEPARTURE (OVALA 1B)			
OVALA 1B DEPARTURE Note: Significant point OVALA at position: 264903.39S 0283744.70E (R170/45DME JSV) 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	21L/R	Left	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 4.5 DME JSV then turn left onto track 111° to intercept R170 JSV (OUBD) to OVALA. Cross 4.5 DME JSV at 7000' ALT or above. Further climb will be under RADAR control. At OVALA set course as per flight plan.

Clearance	RWY	Turn	Details
Communication Failure (OVALA 1B) (Squawk 7600)			
Comply with the OVALA 1B SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At OVALA continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At OVALA route direct to STV and comply with the STANDERTON 5B RWY21L STAR Communication Failure Procedure.			
GRASMERE SIX CHARLIE DEPARTURE (GRASMERE 6C)			
GRASMERE 6C DEP Note: Significant point GAV at position:S263053.63 E0274035.17 5.3% @ 80KT IAS = 429 FPM 5.3% @120KT IAS = 644 FPM 5.3% @160KT IAS = 859 FPM 5.3% @200KT IAS =1073 FPM	21R	Right	Restricted to Turbo Prop aircraft only. Restricted to a minimum climb gradient of 5.3% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to JSV. At JSV turn right onto track 301° (Turn radius to remain within 2.5 DME JSV). Crossing 13 DME JSV turn left direct to GAV. At GAV set course as per flight plan.
Communication Failure (GRASMERE 6C) (Squawk 7600)			
Comply with the GRASMERE SIX CHARLIE SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At GAV set course to the next fix outside 60DME JSV and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At GAV proceed to NIBEX and comply with the NIBEX 2C RWY21L STAR Communication Failure Procedure.			
EXOBI THREE BRAVO DEPARTURE (EXOBI 3B)			
EXOBI 3B DEPARTURE Note: Significant point EXOBI at position: S260252.27 E0290851.32 (R101/50DME JSV) 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	21L/R	Left	Restricted to a minimum climb gradient of 4.2% to the CTR boundary Climb to 8000' ALT, maintain RWY track to 4.5 DME JSV then turn left onto track 111°. Cross 4.5 DME JSV at 7000' ALT or above. Passing 15 DME JSV turn left onto track 076° to intercept R101 JSV (OUBD) to EXOBI. Further climb will be under radar control. At EXOBI set course as per flight plan.
Communication Failure (EXOBI 3B) (Squawk 7600)			
Comply with the EXOBI THREE BRAVO SID, climb to 8300' ALT or maintain last assigned level, whichever is highest. At EXOBI continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At EXOBI proceed to WIV and comply with the WITBANK 3C RWY21L STAR Communication Failure Procedure.			

Clearance	RWY	Turn	Details
HEIDELBERG FIVE DELTA DEPARTURE (HEIDELBERG 5D)			
HEIDELBERG 5D DEPARTURE Note: Significant point HGV at position:S264148.50 E0281659.39 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	21L/R	Left	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to 4.5 DME JSV then turn left direct to HGV. Cross 4.5 DME JSV at 7000' ALT or above. Further climb will be under radar control. At HGV set course as per flight plan.
Communication Failure (HEIDELBERG 5D) (Squawk 7600)			
Comply with the HEIDELBERG FIVE DELTA SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At HGV continue as per flight plan. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At HGV proceed to STV and comply with the STANDERTON 5B RWY21L STAR Communication Failure Procedure.			
LANSERIA ONE CHARLIE DEPARTURE (LANSERIA 1C)			
LANSERIA 1C DEPARTURE Note: Significant point LIV at position:S255656.14 E0275448.86 5.3% @ 80KT IAS = 429 FPM 5.3% @120KT IAS = 644 FPM 5.3% @160KT IAS = 859 FPM 5.3% @200KT IAS =1073 FPM	21R	Right	Restricted to Turbo Prop aircraft only. Restricted to a minimum climb gradient of 5.3% to the CTR boundary. Climb to 8000' ALT, maintain RWY track to JSV. At JSV turn right onto track 301° (Turn radius to remain within 2.5 DME JSV). Crossing 13 DME JSV turn right direct LIV. At LIV intercept R015 LIV (OUBD) to VASUR. At VASUR set course as per flight plan.
Communication Failure (LANSERIA 1C) (Squawk 7600)			
Comply with the LANSERIA ONE CHARLIE SID, climbing to 8300' ALT or maintain last assigned level, whichever is highest. At VASUR set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At VASUR proceed to AVAGO and comply with the AVAGO 2B RW Y21 STAR Communication Failure Procedure.			
EGMEN ONE DELTA RNAV (GNSS) DEPARTURE (EGMEN 1D)			
EGMEN 1D DEPARTURE Note: Significant point EGMEN at position:S253845.74 E0285805.53 (R070/50 DME JSV) 3.8% @140KT IAS = 539 FPM 3.8% @180KT IAS = 693 FPM 3.8% @220KT IAS = 847 FPM	21R	Left	Restricted to a minimum climb gradient of 3.8% to the CTR boundary. Climb to FL090, maintain RWY track 214° to JS008. At JS008 turn left to JS004. At JS004 turn left to JS028. At JS028 turn left to EGMEN. Further climb will be under radar control. At EGMEN set course as per flight plan.Brakpan/Benoni, 5.5 NM SSE, circuit operates at 6000' ALT.

Clearance	RWY	Turn	Details
Communication Failure (EGMEN 1D) (Squawk 7600)			
Comply with the EGMEN ONE DELTA SID, climbing to 8400' ALT or maintain last assigned level, whichever is highest. At EGMEN set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At EGMEN proceed to OKPIT and comply with the OKPIT 4B RWY21L STAR Communication Failure Procedure.			
APDAK ONE DELTA RNAV (GNSS) DEPARTURE (APDAK 1D)			
APDAK 1D DEPARTURE Note: Significant point APDAK at position:S263303.55 E0290252.64 (R137/50 DME JSV) 3.8% @140KT IAS = 539 FPM 3.8% @180KT IAS = 693 FPM 3.8% @220KT IAS = 847 FPM	21R	Left	Restricted to a minimum climb gradient of 3.8% to the CTR boundary Climb to 8000' ALT, maintain RWY track 214° to JS006. At JS006 turn left to JS002. At JS 002 turn right to APDAK. At APDAK set course as per flight plan. Brakpan/Benoni, 5.5 NM SSE, circuit operates at 6000' ALT.
Communication Failure (APDAK 1D) (Squawk 7600)			
Comply with the APDAK ONE DELTA SID, climbing to 8400' ALT or maintain last assigned level, whichever is highest. At APDAK set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At APDAK proceed to STV and comply with the STANDERTON 5B RWY21L STAR Communication Failure Procedure.			
GEROX ONE CHARLIE DEPARTURE (GEROX 1C)			
GEROX 1C DEPARTURE Note: Significant point GEROX at position:S272249.01 E0284943.11 4.2% @140KT IAS = 595 FPM 4.2% @180KT IAS = 766 FPM 4.2% @220KT IAS = 936 FPM	21R	Left	Restricted to a minimum climb gradient of 4.2% to the CTR boundary. Climb to 8000' ALT, maintain RWY track 214° to JS007. At JS007 turn left to GEROX. At GEROX set course as per flight plan.Brakpan/Benoni, 5.5 NM SSE, circuit operates at 6000' ALT.
Communication Failure (GEROX 1C) (Squawk 7600)			
Comply with GEROX ONE CHARLIE SID, climbing to 8400' ALT or maintain last assigned level, whichever is highest. At GEROX set course as per flight plan and climb to flight plan level. Aircraft wishing to return must continue to the SID termination point and climb to last assigned level or MSA if last cleared level is below MSA. At GEROX proceed to STV and enter the STV hold descending to FL240, or if below FL240, hold at last assigned level for minimum 5 MIN. Comply with the STANDERTON 5B RWY21L STAR Communication Failure Procedure.			

12 SPECIAL PROCEDURES

Standard Instrument Departures and Standard Approach Terminal Routes, where in use, are shown on the charts in AD. All SIDs and STARs are subject to change without prior warning, especially during high surface wind and turbulent conditions, to accommodate opposite bound traffic.

OR TAMBO INTERNATIONAL AIRPORT

Applicable Only When Surveillance Radar Is In Operation.

1) Simultaneous use of parallel runways 03L/03R and 21L/21R - O R Tambo International Airport.
2) Simultaneous departures will be permitted, provided it is reasonably sure that radar identification with the respective aircraft will be established within 1 NM of the runway end and:

   a) Tracks will diverge by 15º or more immediately after departure.
   b) 3 NM longitudinal separation can be maintained between the aircraft until a track divergence of 15? or greater is established.

NOTE: Track divergence must be established within 7NM from the airport.

12.1 PARALLEL ILS APPROACHES

NOTE: Only applicable when aircraft are subject to straight-in or radar vectored approaches

a) A minimum of 1000FT vertical separation or minimum 5NM radar separation must be provided between aircraft during the turn-on.
b) A minimum of 5 NM radar separation must be provided between aircraft on the same localiser course (8NM in the case of heavy category aircraft.)
c) A minimum of 3 NM radar separation must be provided between successive aircraft on adjacent localiser courses provided that:-
d) This separation standard will only be applied after the aircraft are established on the respective localisers,
e) The aircraft intend to land off the approaches,

f) Aircraft will be informed when both runways are in use. (This information may be provided through the ATIS.)

12.2 PARALLEL APPROACHES (VISUAL/ILS) OR (VISUAL/ VISUAL)

NOTE: Only applicable when aircraft are subject to straight-in or radar vectored approaches.

1) A minimum of 1000 FT vertical separation or minimum 5NM radar separation will be provided between aircraft during the turn-on.
2) A minimum of 3NM radar separation will be provided between successive aircraft on adjacent final approach tracks
3) This separation will be applied only after the aircraft are established on their respective final approach tracks. (LOC/ VISUAL FINAL - VISUAL FINAL / VISUAL FINAL i.e. both aircraft on visual approaches.)

12.3 CAT II OPERATIONS

During category II operations there will be no parallel approaches unless the standard separations applicable to single runway operations can be maintained.

12.4 PARALLEL ARRIVALS AND DEPARTURES - SAME DIRECTION

1) When conditions dictate the use of one runway for landing and parallel runway for departure the following minima shall apply:-
   a) If the departing aircraft's flight path crosses the extended centre line of the parallel runway, take-off clearance will ensure a minimum of 5NM radar separation from the arriving aircraft.
   b) The departing aircraft's tracks will diverge by 15x or more away from the parallel runway immediately after take-off.
   c) 3NM separation will be maintained between the two aircraft until a track divergence of 15x or more is established between the two aircraft.

12.5 SIMULTANEOUS, OPPOSITE DIRECTION OPERATIONS ON PARALLEL RUNWAYS

1) Simultaneous, opposite direction operations on parallel runways shall be permitted when:-
   a) Conducted in VMC by DAY only: and
   b) When the aircraft operating in the non standard direction is operating under visual flight rules (VFR) and carrying out calibration operations for the Civil Aviation Authority.
   c) When the calibration flight conducts the circuit away from the parallel runway; and
   d) All flights arriving or departing on the runway in use are aware of the calibration operations from the opposite direction.